

Answer Key — Chemistry Grade 10

Sub-Strand 3.1: Acids and Bases • Lesson 4: Differences Between Acids and Bases Using Commercial Indicators

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5
C	B	B	B	D

USE AFTER: OBSERVE PHASE — Experiment 1: recording observations

Q1. Which of these is the most useful observation to record?

- A. "It changed."
- B. "It looked interesting."
- C. "Blue litmus turned bright red within 5 seconds."**
- D. "It was probably an acid."

Correct answer: C. A useful observation states the indicator, the exact colour change, and the timing. Option D is an interpretation, not an observation. Matches the lesson's 'scientist, not a poet' prompt.

USE AFTER: EXPLAIN PHASE — Pattern statements (litmus)

Q2. A few drops of a liquid turn blue litmus paper red. What does this tell you?

- A. The liquid is a base
- B. The liquid is an acid**
- C. The liquid is neutral
- D. The litmus paper is faulty

Correct answer: B. Acids turn blue litmus red. Bases turn red litmus blue. A neutral liquid causes no change to either strip.

USE AFTER: EXPLAIN PHASE — Pattern statements (phenolphthalein)

Q3. In which kind of solution does phenolphthalein turn pink?

- A. An acid
- B. A base**
- C. A neutral solution
- D. Any solution containing water

Correct answer: B. Phenolphthalein is colourless in acids and neutral solutions and turns pink in bases.

USE AFTER: EXPLAIN PHASE — Pattern statements: comparing what each indicator can and cannot show

Q4. A liquid turns methyl orange yellow. Can you be sure it is a base?

- A. Yes, yellow always means a base
- B. No, a neutral solution also turns methyl orange yellow, so test with another indicator**
- C. Yes, but only if the liquid is clear
- D. No, yellow means the liquid is an acid

Correct answer: B. Methyl orange is yellow in both neutral and basic solutions. Red litmus (turns blue only in a base) or phenolphthalein (pink only in a base) is needed to decide.

Q5. Which liquid from today would you expect to turn methyl orange red?

- A. Liquid soap solution
- B. Tap water
- C. Baking soda solution
- D. Vinegar**

Correct answer: D. Methyl orange turns red in acids. Vinegar is an acid. Soap and baking soda are bases (yellow); tap water is neutral (yellow).